Negative Samples Mining Matters: Reconsidering Hyperspectral Image Classification With Contrastive Learning

Hui Liu, Chenjia Huang, Ning Chen, Tao Xie, Mingyue Lu, and Zhou Huang

***Abstract*— In recent years, there have been significant advancements in hyperspectral image (HSI) classification methods using contrastive learning. However, these methods often fail to effectively screen and mine negative samples during the construction of contrastive learning pairs. This oversight introduces negative samples that belong to the same class as the positive samples, thereby limiting the effectiveness of contrastive learning. To address this issue, we propose a novel HSI classification framework based on contrastive learning that flexibly supports the selection and mining of negative samples. Specifically, before training the contrastive learning task, we use pseudolabel information to guide the mining of negative samples, eliminating those with pseudolabels of the same class as the anchor samples. This approach strengthens the alignment between the optimization directions of the contrastive learning task and the classification task. In addition, we improve the contrastive learning process by introducing a controlled mixture of hard and easy negative samples, which enhances the accuracy of HSI classification. The pivotal characteristic of this study lies in enhancing the effectiveness of HSI classification based on contrastive learning through effective filtering and mining of negative samples. Experimental comparisons across multiple public datasets demonstrate the superiority of our proposed method over state-of-the-art algorithms.**

***Index Terms*— Contrastive learning, deep neural network, hyperspectral image (HSI) classification, negative samples mining.**

I. Introduction

HYPERSPECTRAL imaging, as a powerful remote sensing technique, captures the hyperspectral image (HSI) through extremely narrow wavelength ranges across the electromagnetic spectrum, which makes HSI exhibit as a 3-D data cube encompassing hundreds of contiguous spectral bands with high spectral resolution and abundant spatial information [1]. This superiority greatly facilitates the identification and classification of various land-cover types using HSI, that is, HSI classification, aiming at providing the classification map with high accuracy. As for the practical application, precise HSI classification task is of vital significance in the field of environmental monitoring, agricultural resource management, mineral exploration, urban planning, and military reconnaissance. For this purpose, conventional machine learning methods such as support vector machines (SVMs) [2], random forests (RFs) [3], and sparse representations [4] are widely adopted. However, these methods are typically predicated on manual feature extraction, exhibiting limited capability in addressing high-dimensional and nonlinear features and are prone to get trapped into overfitting due to limited training samples [5].

Over the past few years, the explosive growth of deep learning theory and high-performance computing devices has led to significant advancements in hyperspectral image classification based on deep neural networks. Notable representative models include convolutional neural networks (CNNs) [6], deep recurrent neural networks (RNNs) [7], graph convolutional networks (GCNs) [8], stacked autoencoders [9], spatial pyramid pooling [10], fully convolutional networks [11], and other variations such as the diffusion models [12], [13], spectral–spatial residual network (SSRN) [14], deep multiview learning (DMVL) based on ResNet [15], multiscale dense connection fusion networks based on DenseNet [16], [17]. These methods can be specifically designed to automatically and fully extract the spatial–spectral joint features for HSI with high dimension, thereby enhancing adaptability to a wider range of hyperspectral image surface targets in contrast to traditional methods. Although these methods have achieved excellent classification results, they are typically constructed on substantial annotated datasets. Given the fact that obtaining sufficient pixel-level annotation is labor-intensive, time-consuming, and expensive in practice, which is insurmountably challengeable for HSIs related application. This has prompted researchers to explore ways in extracting distinctive spectral–spatial features from limited labeled samples to improve classification accuracy. Consequently, semisupervised learning algorithms have been developed aiming at mining

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limited or few-shot labels. Representative approaches include generative methods [18], such as generative adversarial networks (GANs) [19], variational autoencoder (VAE) [20], and discriminative methods [21], such as self-training, co-training, and graph-based methods [22], [23]. Among these, contrastive learning has recently proven highly effective for learning visual representations in a self-supervised manner, which is becoming an important component of semisupervised learning and is gradually demonstrating its potential in hyperspectral image classification [24].

Contrastive learning trains models by constructing and comparing positive and negative sample pairs generated by an instance discrimination proxy task, enabling the learning of useful feature representations without requiring extensive labeled datasets [25]. The fundamental principle involves maximizing the similarity of positive pairs while minimizing the similarity of negative pairs within the feature space. In recent years, contrastive learning techniques based on deep learning have drawn significant advancements. For instance, Wu initially proposed the instance discrimination task and indicated how to implement unsupervised feature learning using a memory bank [21]. Ye et al. [26] presented unsupervised embedding learning by fully exploited the invariant and spreading instance feature based on fundamental principles of contrastive learning. Afterward, a generative discriminative task, the contrastive predictive coding (CPC), is proposed by maximizing the similarity between the predicted and actual results of sequential data, which also introduced the widely used InfoNCE loss in contrastive learning [27], [28], [29]. Subsequently, several notable models have emerged as powerful contrastive learning techniques, primarily leveraging Siamese neural networks and offering significant methodological advancements. For example, Moco proposed a dictionary lookup technique in addressing the challenge of large sample sizes and consistency [30], [31]; SimCLR used extensive data augmentation methods, projection functions, larger batch sizes, extended training time, and improved loss functions [32], [33]; SwAV adopted the idea of clustering and proposed a multicrop strategy to improve the contrastive learning ability [34]; and BYOL [35], [36] and SimSiam [37] simplify the network and perform contrastive learning without negative samples. In addition, many current contrastive learning tasks use ViT as the backbone network with successful application of transformer and ViT models to enhance feature representation learning [38], [39]. Despite the superior performance of these methods, they are primarily developed for color images with spatial representation in ImageNet classification tasks, its fitness, and effectiveness for limited samples of HSIs with high spectral dimensionality, and the spectral–spatial joint feature representation is still under exploration. Besides, recent research has highlighted the importance of using a large number of negative sample pairs and its effective mining and selection in the contrastive learning process, especially for downstream tasks with different sample quality and quantity [40].

Currently, the HSI-classification-specific contrastive learning improvements have been exploited to fully extract the spectral–spatial features with limited annotated samples, which can be broadly divided into two categories [24], [41], [42], [43], [44]: 1) two-stage models with pretraining and fine-tuning process and 2) the end-to-end models [24]. As for the first category, a typical two-stage training strategy is proposed by Hou to enhance robustness and generalization of feature representations, with unlabeled samples pretrained based on contrastive learning, and the pretrained model is then fine-tuned using a small amount of labeled samples [45]. Similar investigations are conducted by Zhao et al. [46] based on Siamese networks, which mainly concentrated on data augmentation module. Zhu et al. [47] presented a self-supervised contrastive and efficient asymmetric dilated network (SC-EADNet) aiming at learning the multiscale pixelwise contextual feature. Huang et al. [48] proposed a spectral–spatial masked transformer (SS-MTr) based on supervised and contrastive learning, which embedded the local inductive bias. However, most of these two-stage models mainly focused on data augmentation and network architectures, which cannot achieve end-to-end information extraction, and their negative sample generation techniques often result in the separation of similar samples [49]. That is, different instances of the same class may be treated as negative samples, leading to the interference of false-negatives in the contrastive learning process. As for the second category, the CNNs, GCNs, ResNet, and its variations are designed as the end-to-end feature extraction networks with contrastive learning to achieve high-quality feature representations, with representative methods including contrastive GCN (ConGCN), contrastive GCN with skip connection (CGCN-SC), etc. [50]. In addition to transformer and ViT-based network, such as ViT with contrastive learning (CViT) [51], supervised contrastive learning network (SCL-Net) [52], purified contrastive learning (PCL) considering global and local representation [53], and the guided group contrastive learning (GGCL) framework [24]. Although these methods perform well in integrating unsupervised information with supervised signals in an end-to-end manner, the success of contrastive learning highly depends on the design and selection of positive and negative sample pairs, and how to fully leverage negative sample information still remains a significant challenge in the field of unsupervised information extraction [54], [55], [56].

In addition, one of the recent notable studies has highlighted an overlooked aspect for contrastive learning, that is, the impact of hard negatives and easy negatives for unsupervised feature representation, where hard negatives refer to samples that are of a different class but similar to the positive samples, while easy negatives are samples that are of a different class and dissimilar to the positive samples [54]. To circumvent this issue, leading contrastive self-supervised learning methods in computer vision have either significantly increased batch sizes or maintained very large memory banks [57]. However, increased memory requirements lead to diminishing returns in performance. Research has demonstrated that selecting only the hardest negative samples is not optimal [58]. A mixed strategy, incorporating both hard and easy negatives, is generally needed for training [57], [58]. However, the design of this mixed strategy still remains unclear, which we believe may be significantly related to the size and type of the downstream

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task dataset. In this sense, the specific challenges faced by the existing contrastive-learning-based HSI classification tasks include the below.

1) Current contrastive-learning-based HSI classification tasks mainly focus on improvements of data augmentation methods and network architectures, and there is a lack of systematic approaches for flexible selection and extraction of negative samples. Specifically, methods that incorporate prior information for comprehensive negative sample selection and extraction are not yet fully developed [24], [59], [60]. This limitation constrains the potential of contrastive learning to enhance classification accuracy.
2) The existing contrastive learning frameworks typically use random sampling to construct a batch from unlabeled datasets. This can result in uneven class distribution within a batch, increasing the likelihood of false-negative samples and thereby affecting the alignment between the optimization direction of contrastive learning and classification tasks. Therefore, how to effectively integrate prior information to address false-negative samples has become one of the current challenges in contrastive learning.
3) Random negative sampling (RNS) also leads to the imbalance in distribution of hard negatives and easy negatives. However, current contrastive-learning-based HSI classification tasks neglected the impact of hard and easy negatives and lack simple and effective ways to incorporate them. This increases the difficulty in distinguishing between positive and negative samples in contrastive learning, thereby affecting the final performance of hyperspectral classification tasks.

To address these issues, this article proposes a novel framework that leverages contrastive learning to assist HSI classification tasks, which, to the best of our knowledge, is the first to investigate effective negative sample mining for HSI classification. This approach enables flexible selection and mining of negative samples using prior information. Specifically, as false-negatives are challenging to fully resolve due to the lack of labeled information, while semantic information is necessary to learn robust semantic representations. Therefore, to obtain more meaningful negative samples and mitigate the impact of false-negatives, we propose a method that uses pseudolabel information to guide negative sample mining, which specifically consists of two strategies: the **false-negative samples' mitigation (FNSM) strategy** and the **hard–easy negative samples' mixing mining (HENSM) strategy**. The FNSM strategy mitigates the impact of false-negatives and improves the alignment between the optimization directions of contrastive learning and classification tasks. The experimental results show a significant improvement in HSI classification accuracy using this method. In addition, the HENSM strategy was introduced based on our observation that incorporating both hard and easy negative samples in contrastive learning reduces the difficulty in distinguishing between positive and negative samples, leading to better classification outcomes. To address this, we propose a straightforward method for mixing hard and easy negative samples. The experimental results further confirm that this approach enhances HSI classification accuracy. Therefore, we believe that designing specific negative sample mining rules and balancing the hard and easy negatives through thorough design and mining of negative sample information are crucial for contrastive-learning-based HSI classification.

The main contributions of our proposed method are outlined as follows.

1) This article proposes an innovative framework leveraging contrastive learning to assist HSI classification tasks. This framework supports the flexible selection and mining of negative samples required for contrastive learning. Based on this framework, we found that thorough screening and mining of negative samples, according to prior information, can effectively enhance the training performance of contrastive learning, thereby improving the accuracy of HSI classification.
2) This article introduces a novel FNSM strategy to address the learning interference caused by false-negative samples in contrastive-learning-based HSI classification tasks. Specifically, by leveraging pseudolabel information to guide negative sample mining and eliminating samples with the same pseudolabels as the anchor samples, the probability of false-negatives can be significantly reduced. This approach enhances the consistency between the optimization directions of the contrastive learning task and the classification task. The experimental results demonstrate that this method effectively improves the accuracy of HSI classification.
3) This article finds that the difficulty in distinguishing negative samples significantly impacts the training of contrastive learning tasks, thereby affecting classification performance in HSI classification tasks. Based on this observation, we propose a straightforward method called the HENSM strategy, which is used to control the introduction of hard and easy negative samples. The experimental results demonstrate that this method can improve the accuracy of HSI classification.

The rest of this article is structured as follows. Section II addresses the main methodology proposed in this article, including the overall framework, the specific FNSM strategies, and the hard–easy negative mining strategies. Section III presents the detailed experimental setting and the comprehensive results analysis, including the model performance analysis, and ablation studies implemented using three public datasets, followed by the conclusions in Section IV.

II. Methodology

In this section, we will first outline the overall framework of the proposed model. The detailed strategies for the mitigation of false-negative samples, the mining of hard–easy negative samples, and its implementations are presented subsequently.

A. Overview

Our proposed model is a novel contrastive-learning-based HSI classification framework that flexibly supports the selection and mining of negative samples. It consists of two

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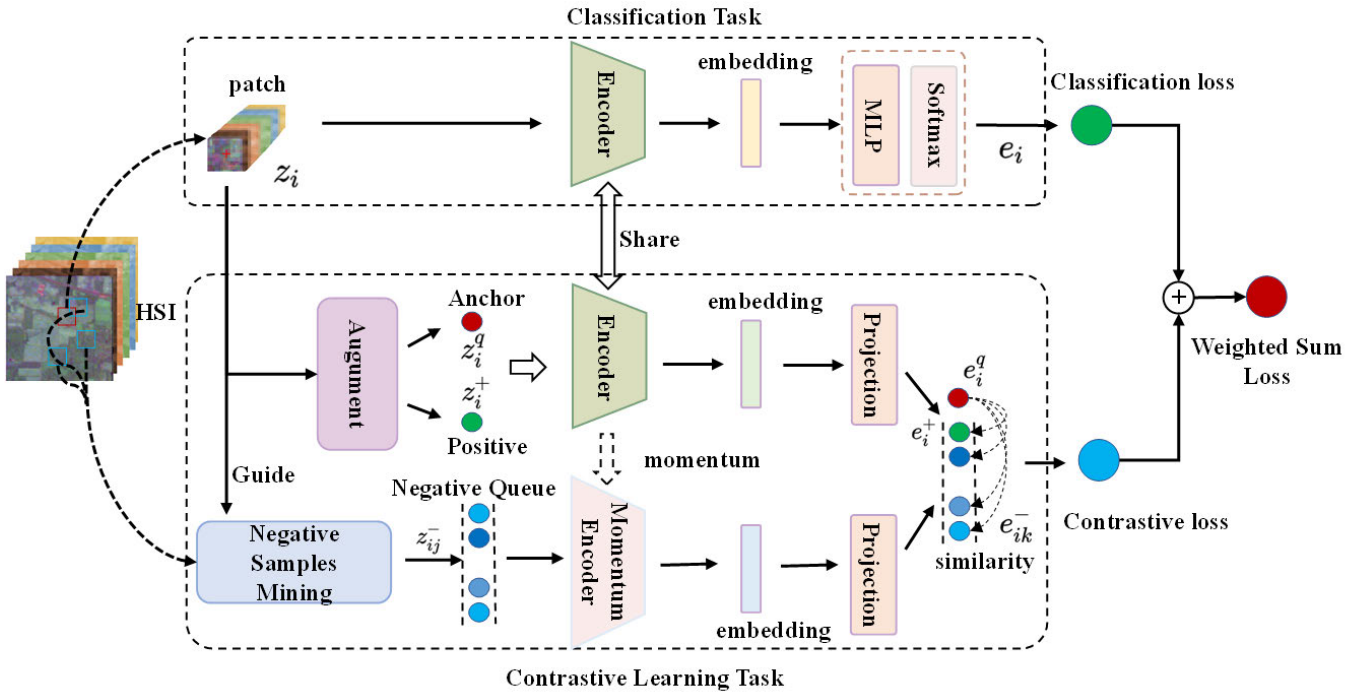


Fig. 1. Framework of our proposed method, which consists of two training tasks: classification task and contrastive learning task.

main components, as illustrated in Fig. 1: the classification architecture and the contrastive learning architecture. These two components can complement and enhance each other, with the contrastive learning task aiding the convergence of the classification model while the classification model generating pseudolabels to assist the contrastive learning task in mining negative samples, ensuring consistency in the optimization directions of both the tasks. The core of our model lies in effective negative sample mining, which uses a momentum encoder to provide a highly adaptive negative sample mining strategy through a negative sample queue technique. Notably, we introduce two key strategies to assist negative sample mining: the **FNSM strategy** and the **HENSM strategy**. Specifically, before training the contrastive learning task, pseudolabels generated by the classification task are used to guide the FNSM and HENSM. The FNSM strategy aims to remove samples whose pseudolabels are of the same class as the anchor samples (i.e., false-negatives), thereby mitigating the sample noise caused by RNS in traditional contrastive learning tasks. This strengthens the consistency in the optimization directions of the contrastive learning and classification tasks. The HENSM strategy aims to address the imbalance in the distribution of hard and easy negative samples, reducing the difficulty in distinguishing positive from negative samples and ensuring better alignment in the optimization directions of the contrastive learning and classification tasks.

As for the overall framework of our proposed method, with the given HSI $\mathcal{H} \in \mathbb{R}^{H\times W\times B}$, where H and W denote the height and width of $\mathcal{H}$, respectively. B represents the number of spectral channels. Assuming the entire HSI dataset consists of N labeled pixels $X_N = \{x_1, x_2, \ldots, x_N\} \in \mathbb{R}^{1\times 1\times B}$ and M unlabeled pixels $X_M = \{x_1, x_2, \ldots, x_M\} \in \mathbb{R}^{1\times 1\times B}$. Each labeled pixel has a class label expressed in one-hot form, $Y_N = \{y_1, y_2, \ldots, y_N\} \in \mathbb{R}^c$, where c represents the number of land-cover classes. Neighboring cubes centered at pixels in X_N and X_M form new datasets $Z_N = \{z_1, z_2, \ldots, z_N\} \in \mathbb{R}^{w\times w\times B}$ and $Z_M = \{z_1, z_2, \ldots, z_M\} \in \mathbb{R}^{w\times w\times B}$, respectively. To effectively leverage the spectral–spatial information comprised in HSIs, the proposed method uses HSI cubes as input. The classification architecture uses a transformer-based encoder as shown in Fig. 2 for feature representation, which is a supervised classifier that trains the labeled dataset X_N to predict pseudolabels for the unlabeled dataset X_M. The specific procedure is that HSI patches are input into the encoder to produce embeddings, which are subsequently processed by the multilayer perceptron (MLP) layer and softmax layers to generate class probabilities P and corresponding pseudolabels for each pixel. Then the cross-entropy (CE) classification loss is adopted and calculated by comparing this probability with the class label, which is described as

$$\mathrm{CE}_{\mathrm{loss}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C}(y_{ij}\log(p_{ij})) \tag{1}$$

where y_{ij} is the true label, and $p_{ij} \in P$ is the predicted probability for category j of sample i. N and C represent the number of samples and categories, respectively.

In terms of the contrastive learning task architecture, the objective is to learn a transformation that brings positive sample pairs closer in the embedding space while pushing negative sample pairs far apart. The process involves the following steps.

1) *Data Augmentation:* For each batch of data, augmentation of the spectral odd–even mask is applied to generate different views of the same sample.

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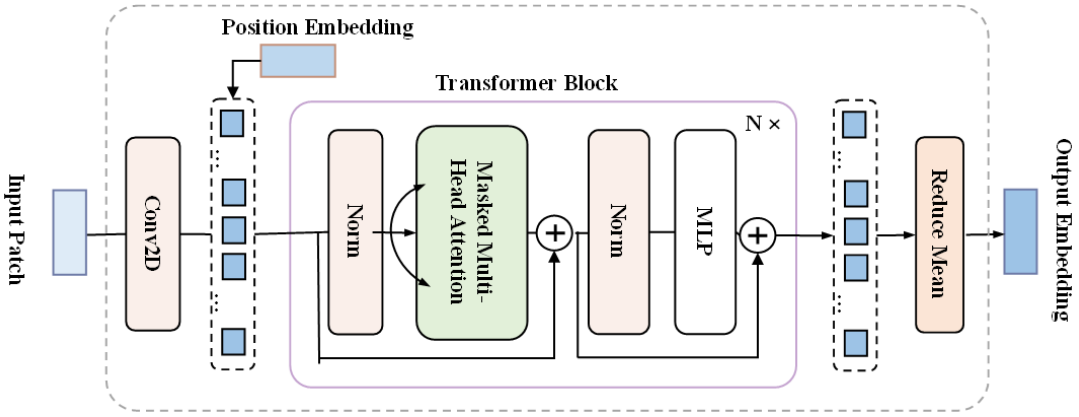


Fig. 2. Detailed structure of the encoder block.

2) *Negative Samples Mining Module:* As mentioned previously, the selection of negative samples is crucial in this process, as it determines whether the negative samples are informative and beneficial for the feature learning process. In this module, we use two negative sample selection strategies: FNSM strategy and HENSM strategy. To intuitively illustrate the proposed strategies and compare it with negative sampling mode in traditional contrastive learning models, we demonstrate the details of negative mining strategies in Fig. 3. Fig. 3(a) illustrates that traditional contrastive learning typically uses RNS, where the samples are unlabeled and randomly distributed; Fig. 3(b) denotes that by introducing the FNSM strategy in this study, each sample is assigned with pseudolabels generated by the classifier, allowing for the identification of false-negative samples; Fig. 3(c) shows the elimination of false-negatives based on the FNSM strategy; Fig. 3(d) depicts the calculation of the spatial–spectral joint similarity between anchor and negative samples, which is used to determine the hardness level of the negative samples, In Fig. 3(e), negative samples are ranked by hardness level, and hard–easy negative sample mixing is performed by controlling this hardness level; Fig. 3(f) demonstrates contrastive learning, where similar samples are pulled closer and dissimilar samples are pushed farther apart. Specific explanations of these strategies will be provided in Sections II-B and II-C.
3) *Shared Encoder:* The encoder used for positive samples is identical to the classification encoder and shares parameters with it.
4) *Momentum Encoder for Negatives:* To enhance the utilization of negative samples, a momentum encoder is used for learning the features of negative samples. This encoder maintains a moving average of the encoder parameters to provide more stable and consistent negative representations.
5) *Projection:* The final embedding features are obtained through a projection head. This helps in refining the embeddings before calculating the contrastive loss.

The InfoNCE loss is used as our contrastive loss function. For a positive anchor sample with its embedding e_i^q, its positive match e_i^+, and a negative samples e_{ik}^- from the queue,

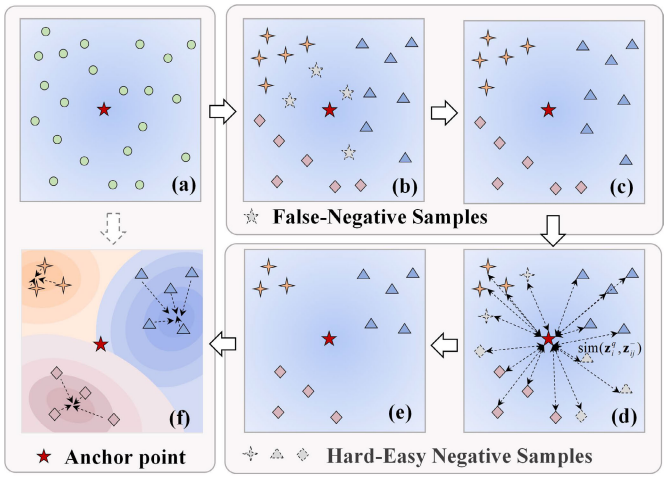


Fig. 3. Details of the negative samples' mining strategies. (a) RNS in traditional contrastive learning, where samples are unlabeled and randomly distributed. (b) Pseudolabels generated by the classifier are assigned to each sample, where false-negative samples are identified. (c) False-negative samples are eliminated based on FNSM strategy. (d) The spatial–spectral joint similarity between the anchor and negative samples is calculated to represent the hardness level of the negative samples. (e) HENSM is conducted by controlling the hardness level of the negative samples. (f) Contrastive learning adjusts the representation by pulling similar samples closer and pushing dissimilar samples farther apart. In this framework, to facilitate the comparisons, (a) and (f) illustrate traditional RNS in contrastive learning, (b) and (c) demonstrate the FNSM strategy, while (d) and (e) represent the HENSM strategy.

where $k \in \{1, \ldots, K\}$, the loss is defined as follows:

$$\text{InfoNCE}_{\text{loss}} = -\log \frac{\exp\left(\text{sim}\left(e_i^q, e_i^+\right)/\tau\right)}{\sum_{k=1}^{K} \mathbb{1}_{[k\neq i]} \exp\left(\text{sim}\left(e_i^q, e_{ik}^-\right)/\tau\right)} \quad (2)$$

where K represents the number of samples in the negative queue, $\mathbb{1}_{[k\neq i]} \in \{0, 1\}$ is an indicator function that equals 1 if $k \neq i$, and $\text{sim}(e_i^q, e_i^+) = (e_i^q \cdot e_i^+)/(\|e_i^q\|\|e_i^+\|)$ represents the cosine similarity between two vectors e_i^q and e_i^+. The parameter τ denotes a temperature parameter, with a default value of 0.5. The total loss is computed across all the positive pairs within a mini-batch.

The contrastive training process involves minimizing the InfoNCE loss, which ensures that positive samples are brought closer in the embedding space while negative samples are pushed farther apart. The final model training is guided by a weighted sum of the classification loss and the contrastive

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loss, enabling the model to leverage both supervised and unsupervised learning signals effectively, which is defined as

$$\text{Loss} = \text{CE}_{\text{loss}} + \lambda \cdot \text{InfoNCE}_{\text{loss}} \tag{3}$$

where λ controls the weight of InfoNCE loss. In this way, the proposed method integrates a classification network with a contrastive learning network. The classification network provides pseudolabels, while the contrastive learning network refines the embeddings by learning to distinguish between positive and negative pairs. This dual-guidance approach optimizes the model for hyperspectral image classification by balancing the utilization of both easy and hard negative samples.

B. False-Negative Samples' Mitigation

Due to the absence of supervisory information, the existing contrastive learning frameworks typically construct batches of negative sample pairs through random sampling. However, randomly sampled negative samples may introduce false-negatives, thereby resulting in performance degradation. Specifically, during the construction of positive and negative sample pairs in contrastive learning, some negative samples z_{ij}^-, where $j \in \{1, \ldots, m\}$, might actually belong to the same class as the anchor point z_i^q but are incorrectly identified as negative samples due to differences in certain features, such as spectral characteristics. In the context of contrastive-learning-based HSI classification, such samples are referred to as **false-negative samples**. The presence of false-negative samples can lead to a misalignment between the optimization direction of the contrastive learning task and the classification task, thereby slowing down the model's convergence and negatively impacting the classification results.

To mitigate the impact of false-negative samples, it is crucial to effectively use sample features and accurately identify these samples. Therefore, we propose a pseudolabel-guided strategy to eliminate the false-negatives. Specifically, a base classifier is first trained to assign pseudolabels to each sample. Once an anchor point is selected, negative samples can be categorized into those with the same label as the anchor point and those with different labels, the former are more likely to contain a great number of false-negatives. Based on this categorization, we aim to eliminate false-negative samples from the negative sample set as effectively as possible. In practice, we implement the following training strategy:

1) Based on the labeled data Z_N, we initially train the classification tower in the network to endow the classification structure with preliminary classification capabilities.
2) Using the classification tower, we assign pseudolabels to the unlabeled data Z_M.
3) For anchor point z_i^q, obtain a set of pseudolabeled samples F_i, then remove samples with the same labels with z_i^q, obtain negative samples set N_i, and use this set as the negative sample set for contrastive task training.

We found that by leveraging pseudolabel information to guide negative sample mining and eliminating samples with the same pseudolabels as the anchor point, the occurrence of false-negative samples can be significantly reduced. This approach enhances the consistency between the optimization directions of the contrastive learning task and the classification task, thereby effectively improving the accuracy of HSI classification.

C. Hard–Easy Negative Samples' Mixing Mining

Having eliminated the interference of false-negatives, the selection of truly informative negative samples becomes crucial for effective feature representation in contrastive learning. In the existing contrastive learning frameworks, negative samples z_{ij}^- are typically randomly sampled from the training data for a given anchor z_i^q. The information content of these negative samples, and thus their effectiveness in the learning process, remains unknown. Research indicates that effective true negatives can expedite the correction of errors in supervised and metric learning. However, introducing an excessive number of hard negative samples is suboptimal; a mixture of hard and easy negative samples is typically required for training [57], [58]. Therefore, to effectively identify more informative negative samples, we first outline the characteristics of hard and easy negative samples as guiding principles:

1) *Hard Negative Samples:* z_{ij}^- are true negatives with different labels from the anchor point z_i^q (i.e., they belong to different classes) but possess similar features, making them difficult to distinguish from the anchor point.
2) *Easy Negative Samples:* z_{ij}^- are true negatives with different labels from the anchor point z_i^q while possess dissimilar features, making them easy to distinguish from the anchor point.

In representation learning, informative negative samples are intuitively those that are mapped close to the anchor but should be far apart. In other words, negative samples with different labels from the anchor but embedded nearby are likely to be the most useful, providing significant gradient information during training. This concept has been successfully applied in metric learning, where truly dissimilar sample pairs are known and accessible, enabling the acquisition of true negative pairs [54]. This contrasts with unsupervised contrastive learning, where such supervision is absent. Given the lack of supervision in unsupervised contrastive learning, we propose an efficient and easily implementable negative sample mining strategy guided by pseudolabel information. With the guidance of pseudolabels, negative samples with different labels from the anchor can be effectively identified. In addition, our method allows for the adjustment of the "hardness" level of negative samples based on spatial–spectral similarity. This flexibility enables users to balance the selection of optimal negative samples from hard and easy negatives. Specifically, our proposed pseudolabel-guided NENSM method is as follows.

1) Having mitigated the false-negatives, now the negative samples' set N_i can be obtained for each anchor point z_i^q, where N_i primarily includes the true negatives.
2) Compute the spatial–spectral joint similarity between the anchor point z_i^q and each negative sample z_{ij}^- in N_i,

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the similarity score is defined as score $=$ sim(z_i^q, z_{ij}^-). The similarity function sim is a weighted combination of spatial similarity spa and spectral similarity spe. The formula for this calculation is as follows:

$$\text{score} = e^{w\times\text{spe}+(1-w)\times\text{spa}} \tag{4}$$

$$\text{spe} = \arccos\left(\frac{z_i^q \cdot z_{ij}^-}{\|z_i^q\| \cdot \|z_{ij}^-\|}\right) \tag{5}$$

$$\text{spa} = \sqrt{\left(W_{z_i^q} - W_{z_{ij}^-}\right)^2 + \left(H_{z_i^q} - H_{z_{ij}^-}\right)^2}. \tag{6}$$

Here, $W_{z_i^q}$ and $W_{z_{ij}^-}$ represent the x-coordinates of the anchor and the negative sample, respectively, while $H_{z_i^q}$ and $H_{z_{ij}^-}$ represent the y-coordinates. The parameter w denotes the weight for spatial similarity with the range between 0 and 1. The spa and spe are normalized within the range of 0–1 to ensure the balanced impact.

3) Introduce the negative sample number K as a hyperparameter and sort the negative samples in N_i by similarity. Hence, the hyperparameter K can control the hardness of negative samples by specifying the number of negative samples to extract based on their similarity scores.

D. Implementation

The proposed method enhances HSI classification through a contrastive-learning-assisted classification framework, which involves training a classifier to obtain pseudolabels for samples, eliminating false-negatives, and effectively mining hard and easy negative samples under the guidance of pseudolabel information. The specific implementation process with pseudocode is illustrated in Algorithm 1. As is shown, a supervised classifier is first trained on labeled data to predict pseudolabels for the unlabeled data, forming a pseudolabeled dataset. Then, the false-negatives are identified and excluded, ensuring true negatives dominate the sample set. Negative samples are further refined by computing the spatial–spectral joint similarity and the negative sample numbers, which is used to control the hardness level of negative samples. The model is then trained using these refined negative samples, adjusting the number of negative samples to balance the inclusion of hard and easy negatives, which optimizes feature learning and prevents overfitting.

Algorithm 1 Contrastive Learning With False-Negative Mitigation and Mixed Negative Sample Mining

Input: Training dataset D, threshold K
Output: Trained model

Phase 1: Pseudolabel Generation
Train a classifier Ψ on D to obtain pseudolabels for all samples.
Phase 2: False-Negative Mitigation
for each anchor point z_i^q in D **do**
  Get the set of negative samples F_i for anchor z_i^q.
  Use Ψ to assign pseudolabels for F_i.
  Identify potential false-negatives which have the same class with the anchor sample z_i^q.
  Remove false-negatives from F_i, and obtain negative samples' set N_i.
end for
Phase 3: Hard–Easy Negative Sample Mixing Mining
for each anchor point z_i^q in D **do**
  Calculate the cosine similarity between the anchor z_i^q and each true negative sample $z_{ij}^- \in N_i$.
  Ranking negative samples from hard to easy based on similarity scores.
end for
Phase 4: Co-Training
for each training epoch **do**
  for each batch of data **do**
    Perform data augmentation to generate positive pairs.
    Compute embeddings for anchor points, positive pairs, and negative samples.
    Calculate contrastive loss and classification loss.
    Update the model parameters by minimizing the weighted sum of classification loss and contrastive loss.
  end for
end for

III. Experiments

A. Experimental Settings

1) Datasets: To verify the effectiveness of our algorithm, we applied the algorithm to three public datasets: the Indian Pines dataset (IP), the WHU-Hi-HongHu dataset (WHHH), and the Salinas(SA) dataset.

The Indian Pines dataset, acquired in 1992 using the airborne visible/infrared imaging spectrometer (AVIRIS) sensor, encompasses the northwestern area of Indiana, USA. This dataset consists of 224 spectral bands spanning wavelengths from 0.4 to 2.5 μm, with a spatial resolution of 20 m across 145 $\times$ 145 pixels. It incorporates 16 distinct land-cover classes. For our analysis, we removed 24 bands related to water absorption and noise, thus narrowing the selection to 200 bands. The false-color image and the classification ground truth are illustrated in Fig. 4.

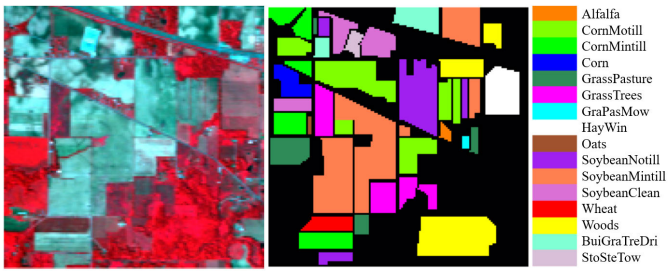


Fig. 4. False-color composite image, the corresponding ground-truth map, and the legend of the Indian Pines dataset.

The WHU-Hi-HongHu dataset [61], [62] was obtained on November 20, 2017, between 16:23 and 17:37 in Honghu City, Hubei Province, China. This dataset was captured using a Headwall Nano-Hyperspec imaging sensor with a 17-mm focal length, mounted on a DJI Matrice 600 Pro unmanned aerial vehicle (UAV) operating at an altitude of 100 m. The

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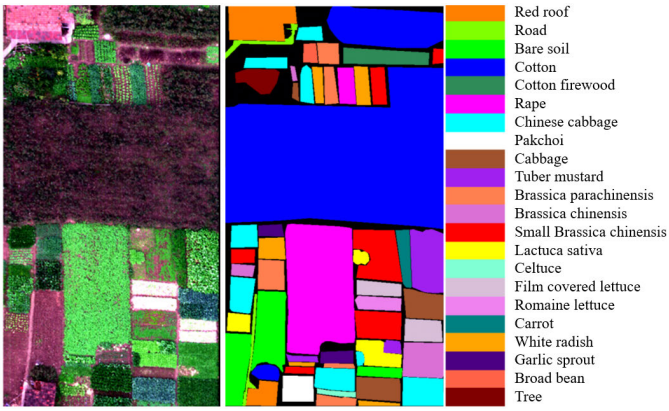


Fig. 5. False-color composite image, the corresponding ground-truth map, and the legend of the WHU-Hi-HongHu dataset.

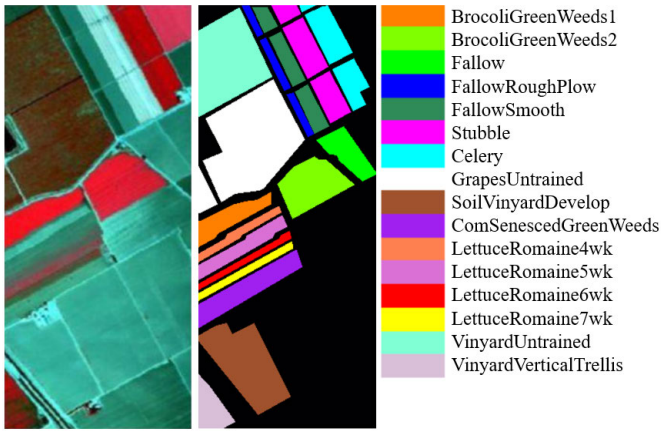


Fig. 6. False-color composite image, the corresponding ground-truth map, and the legend of the Salinas dataset.

resulting images, with a resolution of 940 × 475 pixels, encompass 270 spectral bands ranging from 400 to 1000 nm, offering a spatial resolution of approximately 0.043 m for the hyperspectral UAV imagery. The captured scene represents a complex agricultural landscape, featuring various crop types, including different varieties of the same crop, such as Chinese cabbage, cabbage, Brassica chinensis, and small Brassica chinensis. In total, the dataset covers 22 distinct land-cover classes. The false-color image and the classification ground truth are illustrated in Fig. 5.

The Salinas dataset, collected by an airborne hyperspectral sensor, covers an agricultural region in Salinas Valley, California, USA. This dataset comprises 512 × 217 pixels with 224 spectral bands spanning wavelengths from 400 to 2500 nm and features a spatial resolution of approximately 3.7 m. It includes 16 distinct crop types, such as lettuce, broccoli, onion, and cauliflower. The Salinas dataset has been extensively used in classification, clustering, and feature extraction studies. Its availability has significantly contributed to advancements in hyperspectral imaging and agricultural monitoring research. The false-color image and the classification ground truth are depicted in Fig. 6.

2) Evaluation Metrics: The effectiveness of our algorithm is evaluated and compared with other algorithms based on four indicators, primarily including overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and the classification accuracy for each land-cover category.

TABLE I
TRAINING AND TEST SAMPLE NUMBERS IN THE INDIAN PINES DATASET

ID	Class	Train.	Test.	Total
0	Alfalfa	10	36	46
1	CornMotill	10	1418	1428
2	CornMintill	10	820	830
3	Corn	10	227	237
4	GrassPasture	10	473	483
5	GrassTrees	10	720	730
6	GrassPastureMowed	10	18	28
7	HayWindrowed	10	468	478
8	Oats	10	10	20
9	SoybeanNotill	10	962	972
10	SoybeanMintill	10	2445	2455
11	SoybeanClean	10	583	593
12	Wheat	10	195	205
13	Woods	10	1255	1265
14	BuildingsGrassTreesDrives	10	376	386
15	StoneSteelTowers	10	83	93
	Total	160	10089	10249

3) Training Details: The PyTorch training framework was used for the implementation and training of the model, with the basic hardware environment comprising an Intel[1] Xeon[1] Platinum 8336C CPU at 2.30 GHz, 64 cores, 256 GB of memory, and two NVIDIA GeForce RTX 4090 GPUs, each with 24 GB of memory. The division of training and testing datasets used in our study is shown in Tables I–III.

4) Parameter Analysis: To comprehensively evaluate the effect of hyperparameters on different datasets, we conducted a hyperparameter analysis experiment. Specifically, we tested two parameters of weight w and the number of negative samples K on the IP, WHHH, and SA datasets. The weight controls the spatial similarity in the spatial–spectral joint similarity calculation, while the number of negative samples in the contrastive learning queue directly affects the proportion of hard negative samples, a key factor influencing the effectiveness of contrastive learning. As shown in Fig. 7, the optimal control weight for achieving the highest OA varies across different datasets. The IP and SA datasets perform best with a lower weight value of 0.2, whereas the WHHH dataset benefits from a moderate weight value of 0.4. In addition, Fig. 8 evaluates the impact of the chosen number of negative samples on OA within the same dataset. It is evident that the optimal number of negative samples also differs across datasets. Theoretically, increasing the number of negative samples should improve classification performance. However, our experiments indicate that beyond a certain point, increasing the number of negative samples does not lead to better results. This is because a larger number of negative samples increase the proportion of hard negative samples, which can be detrimental to classification performance. As shown in Fig. 8, the IP

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TABLE II
TRAINING AND TEST SAMPLE NUMBERS IN THE WHU-HI-HONGHU DATASET

ID	Class	Train.	Test.	Total
0	Red roof	10	14031	14041
1	Road	10	3502	3512
2	Bare soil	10	21811	21821
3	Cotton	10	163275	163285
4	Cotton firewood	10	6208	6218
5	Rape	10	44547	44557
6	Chinese cabbage	10	24093	24103
7	Pakchoi	10	4044	4054
8	Cabbage	10	10809	10819
9	Tuber mustard	10	12384	12394
10	Brassica parachinensis	10	11005	11015
11	Brassica chinensis	10	8944	8954
12	Small Brassica chinensis	10	22497	22507
13	Lactuca sativa	10	7346	7356
14	Celtuce	10	992	1002
15	Film covered lettuce	10	7252	7262
16	Romaine lettuce	10	3000	3010
17	Carrot	10	3207	3217
18	White radish	10	8702	8712
19	Garlic sprout	10	3476	3486
20	Broad bean	10	1318	1328
21	Tree	10	4030	4040
	Total	220	386473	386693

TABLE III
TRAINING AND TEST SAMPLE NUMBERS IN THE SALINAS DATASET

ID	Class	Train.	Test.	Total
0	BrocoliGreenWeeds1	10	1999	2009
1	BrocoliGreenWeeds2	10	3716	3726
2	Fallow	10	1966	1976
3	FallowRoughPlow	10	1384	1394
4	FallowSmooth	10	2668	2678
5	Stubble	10	3949	3959
6	Celery	10	3569	3579
7	GrapesUntrained	10	11261	11271
8	SoilVinyardDevelop	10	6193	6203
9	ComSenescedGreenWeeds	10	3268	3278
10	LettuceRomaine4wk	10	1058	1068
11	LettuceRomaine5wk	10	1917	1927
12	LettuceRomaine6wk	10	906	916
13	LettuceRomaine7wk	10	1060	1070
14	VinyardUntrained	10	7258	7268
15	VinyardVerticalTrellis	10	1797	1807
	Total	160	53969	54129

and WHHH datasets achieve the best results with a moderate number of 256 negative samples. This finding underscores the importance of mixing hard and easy negatives for optimal performance tailored to specific datasets.

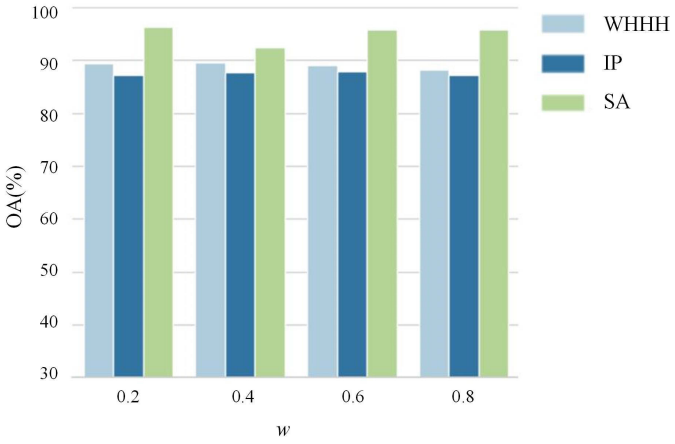


Fig. 7. Classification performance for the IP, WHHH, and SA datasets under different weights.

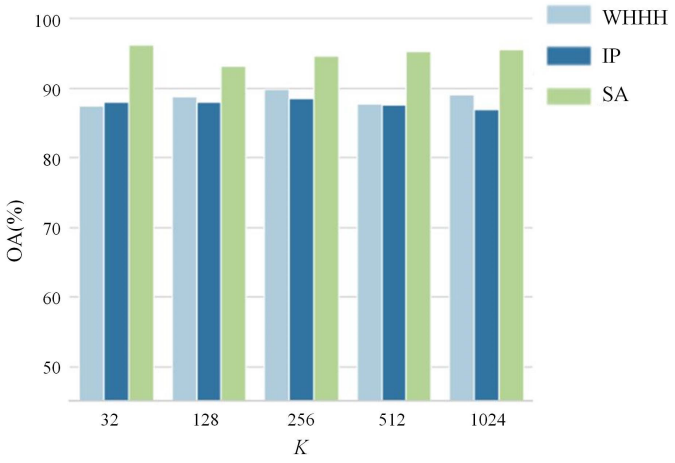


Fig. 8. Classification performance for the IP, WHHH, and SA datasets under different number of negative samples.

B. Performance Analysis

To evaluate the effectiveness of the proposed method, we conducted comparative experiments using a variety of representative algorithms. These methods include HiT, SSFTT, CASST, SC-EADNet, S3Net, and SS-MTr, aiming to benchmark the proposed method against the existing algorithms. The descriptions of these algorithms are as follows.

1) *HiT:* Hyperspectral image transformer (HiT) [63] is based on ViT framework, embedding convolution operations within ViT to handle joint spectral–spatial features. This method represents a prominent transformer-based algorithm for HSI classification tasks. By integrating convolutional operations, HiT leverages both spatial and spectral information, enhancing the model's ability to capture complex patterns in HSI data.
2) *SSFTT:* Spectral–spatial feature tokenization transformer (SSFTT) [64] effectively combines CNNs and transformers, introducing a Gaussian-weighted feature tokenizer to further improve model performance. It stands as one of the competitive algorithms in current transformer-based HSI classification tasks. The integration of CNN and transformer components enables SSFTT to benefit from the localized feature extraction

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TABLE IV
CLASSIFICATION RESULTS OF EXPERIMENTS ON THE INDIAN PINES DATASET (TEN LABELS PER CLASS)

Class	HiT	SSFTT	CASST	SC-EADNet	S3Net	C-SS-MTr	Ours
1	99.44 ± 1.24	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	98.33 ± 1.52	98.44 ± 1.07	$\mathbf{100.00 \pm 0.00}$
2	40.42 ± 11.73	75.98 ± 2.67	78.03 ± 2.30	70.58 ± 2.48	62.33 ± 4.42	58.59 ± 1.38	$\mathbf{91.44 \pm 2.05}$
3	31.17 ± 6.67	84.63 ± 1.58	79.54 ± 2.04	73.20 ± 2.95	$\mathbf{84.83 \pm 5.90}$	69.49 ± 2.36	75.02 ± 2.42
4	63.35 ± 24.62	93.30 ± 3.83	91.28 ± 1.93	95.51 ± 3.58	98.59 ± 0.85	96.00 ± 2.50	$\mathbf{99.82 \pm 0.39}$
5	38.09 ± 4.69	$\mathbf{94.71 \pm 1.11}$	80.93 ± 0.59	84.65 ± 0.12	92.30 ± 0.85	78.65 ± 2.19	93.02 ± 1.82
6	73.22 ± 9.47	93.36 ± 0.58	97.89 ± 0.54	91.44 ± 1.12	$\mathbf{98.03 \pm 1.12}$	85.56 ± 1.76	94.97 ± 1.41
7	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	98.67 ± 1.45	$\mathbf{100.00 \pm 0.00}$
8	98.58 ± 1.31	$\mathbf{100.00 \pm 0.00}$	99.87 ± 0.19	97.82 ± 1.15	99.70 ± 0.42	99.64 ± 0.50	87.26 ± 8.10
9	98.00 ± 4.47	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$
10	60.09 ± 16.08	79.83 ± 3.65	75.86 ± 2.60	78.15 ± 3.54	78.09 ± 7.07	76.77 ± 2.29	$\mathbf{86.59 \pm 1.76}$
11	56.92 ± 9.22	67.46 ± 4.00	77.98 ± 1.58	78.38 ± 5.58	71.75 ± 3.34	71.20 ± 1.37	$\mathbf{78.53 \pm 4.08}$
12	32.71 ± 10.14	$\mathbf{81.37 \pm 6.01}$	69.43 ± 4.16	70.05 ± 9.61	72.11 ± 4.72	67.05 ± 2.77	79.14 ± 3.75
13	81.54 ± 5.54	$\mathbf{100.00 \pm 0.00}$	98.05 ± 1.83	99.90 ± 0.23	99.28 ± 1.61	96.10 ± 2.01	99.38 ± 0.67
14	84.01 ± 16.40	97.91 ± 0.51	87.59 ± 2.98	93.00 ± 2.41	98.06 ± 2.69	88.58 ± 2.23	$\mathbf{99.47 \pm 0.20}$
15	44.30 ± 8.24	98.35 ± 0.55	98.35 ± 1.86	97.93 ± 1.29	91.70 ± 1.84	84.49 ± 1.13	$\mathbf{98.72 \pm 1.86}$
16	93.49 ± 4.48	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	99.52 ± 0.66	97.59 ± 0.85	95.90 ± 1.23	$\mathbf{100.00 \pm 0.00}$
OA(%)	58.79 ± 4.74	83.30 ± 1.35	82.95 ± 0.41	82.20 ± 1.03	81.79 ± 1.28	76.19 ± 0.31	$\mathbf{87.67 \pm 0.95}$
AA(%)	68.46 ± 3.02	91.68 ± 0.63	89.67 ± 0.36	89.38 ± 0.42	90.17 ± 0.84	85.30 ± 0.36	$\mathbf{92.71 \pm 0.71}$
$\kappa \times 100$	53.24 ± 5.16	81.20 ± 1.49	80.72 ± 0.48	79.96 ± 1.12	79.36 ± 1.42	73.25 ± 0.35	$\mathbf{86.06 \pm 1.05}$

of CNNs and the global context understanding of transformers.

3) *CASST:* Spatial–spectral transformer with cross-attention (CASST) [65] uses a cross-attention module to effectively integrate spectral and spatial information, making it another competitive algorithm in the realm of transformer-based HSI classification tasks. The cross-attention mechanism in CASST allows the model to align and fuse spectral and spatial features, enhancing its discriminatory power for HSI data.
4) *SC-EADNet:* Spatial–spectral transformer with cross-attention (SC-EADNet) [47] uses a self-supervised multiscale feature learning model to fully exploit per-pixel contextual information. It uses asymmetric dilated convolution blocks within a contrastive learning framework to achieve competitive classification performance with rapid computational speed. This method emphasizes efficient feature extraction and contrastive learning to improve model robustness and accuracy.
5) *S3Net:* Spectral–spatial Siamese network (S3Net) [66] is a coherent end-to-end contrastive learning model designed for few-shot HSI classification. It uses weighted contrastive loss and adaptive CELoss, using a limited amount of labeled data while effectively mitigating overfitting. The Siamese network architecture of S3Net enables it to learn discriminative features even from a small number of training samples, making it suitable for scenarios with limited labeled data.
6) *C-SS-MTr:* Spectral–spatial masked transformer (C-SS-MTr) [48] is a typical two-stage training model based on contrastive learning. This method considers local inductive bias within the transformer architecture, providing strong instance discrimination capabilities. The masked transformer approach in C-SS-MTr allows it to focus on relevant spectral–spatial features while ignoring irrelevant background information, enhancing its classification accuracy.

1) Quantitative Results: To quantitatively analyze the effectiveness of our proposed method, we aggregated the classification performance on the IP, WHHH, and SA datasets, as shown in Tables IV–VI, respectively. Each table includes the classification accuracy for each method across different classes, as well as OA, AA, and Kappa coefficient (κ). The best performance for each metric is highlighted in bold.

As indicated in Tables IV–VI, under the condition of using ten samples per class as the training set, the HiT model exhibits lower performance compared with other algorithms across all three datasets. The SC-EADNet model also shows relatively low performance on the WHHH dataset, indicating that there exist further improvements for limited training samples. Moreover, except for the HiT model, other models exhibit good classification accuracy on the SA dataset, indicating their strong classification capability in limited sample training scenarios. Notably, compared with other algorithms, our proposed method further improves the classification performance, with OA, AA ,and κ all achieving the best results. Specifically, compared with the second best method, our method improves OA by 4.37%, 0.73%, and 0.78% on the IP, PU and WHHH datasets, respectively. In addition, AA and K are also enhanced by 1.03%, 0.49%, and 0.35%, and 4.86%, 0.88%, and 0.86% on these three datasets, respectively.

When observing the accuracy of each subclass in each dataset, we find that our proposed method still outperforms the compared models. As shown in Table IV, the classification accuracy of ten of the 16 subclasses in the IP dataset is significantly higher than other methods, especially for the second and tenth subclasses compared with the second best methods CASST and SSFTT, the method improved by 13.41% and 6.76%, respectively. For subclasses of the IP dataset where our algorithm performs relatively poor, the performance difference is relatively small compared with the highest performing classification metric in that subclass. As can be seen in Table V, our method also performs well in the subsample classification of the WHHH dataset. For example, the 11th,

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TABLE V
CLASSIFICATION RESULTS OF EXPERIMENTS ON THE WHU-HI-HONGHU DATASET (TEN LABELS PER CLASS)

Class	HiT	SSFTT	CASST	SC-EADNet	S3Net	C-SS-MTr	Ours
1	93.93 ± 1.16	91.93 ± 1.04	89.21 ± 0.50	$\mathbf{95.29 \pm 1.46}$	91.20 ± 3.29	92.92 ± 1.19	89.86 ± 3.06
2	73.79 ± 6.87	81.52 ± 4.43	83.98 ± 3.25	76.08 ± 7.37	74.04 ± 4.99	$\mathbf{88.68 \pm 1.11}$	85.25 ± 5.78
3	78.99 ± 3.86	80.72 ± 0.68	81.83 ± 1.68	69.20 ± 2.43	$\mathbf{86.80 \pm 4.54}$	76.86 ± 1.23	86.52 ± 2.42
4	92.16 ± 0.75	90.73 ± 1.06	94.96 ± 1.35	91.31 ± 1.51	90.12 ± 5.66	$\mathbf{98.40 \pm 0.38}$	97.45 ± 0.72
5	86.60 ± 2.55	92.68 ± 2.32	94.14 ± 1.63	48.99 ± 6.36	75.15 ± 9.62	82.52 ± 2.19	$\mathbf{96.75 \pm 0.49}$
6	83.80 ± 2.79	91.84 ± 0.74	93.25 ± 1.27	81.27 ± 1.47	87.94 ± 3.86	90.97 ± 0.74	$\mathbf{95.89 \pm 1.11}$
7	50.72 ± 3.38	66.44 ± 2.71	68.84 ± 2.60	42.89 ± 4.10	66.76 ± 8.87	70.09 ± 5.00	$\mathbf{71.56 \pm 2.42}$
8	33.90 ± 15.23	65.44 ± 3.07	79.10 ± 2.94	33.36 ± 2.25	47.93 ± 12.89	71.56 ± 2.21	$\mathbf{80.24 \pm 4.96}$
9	82.86 ± 8.08	93.85 ± 0.87	95.39 ± 1.30	60.38 ± 1.56	$\mathbf{96.28 \pm 1.99}$	94.11 ± 1.51	94.51 ± 1.55
10	33.34 ± 9.86	70.90 ± 7.99	$\mathbf{81.53 \pm 3.65}$	42.22 ± 5.49	62.15 ± 12.30	80.10 ± 2.66	76.31 ± 3.92
11	23.10 ± 5.77	61.42 ± 3.28	66.82 ± 1.65	32.36 ± 11.32	63.60 ± 7.31	63.97 ± 1.11	$\mathbf{71.56 \pm 6.66}$
12	60.77 ± 5.87	58.37 ± 5.38	66.96 ± 1.29	53.11 ± 3.80	49.02 ± 9.70	73.80 ± 3.67	$\mathbf{80.83 \pm 3.35}$
13	47.01 ± 5.83	48.20 ± 7.27	$\mathbf{76.09 \pm 3.49}$	39.25 ± 5.72	50.07 ± 2.60	72.03 ± 3.64	54.15 ± 7.80
14	60.74 ± 5.73	92.56 ± 2.38	$\mathbf{92.68 \pm 1.46}$	56.24 ± 1.54	85.52 ± 5.51	68.11 ± 2.47	92.33 ± 2.61
15	92.24 ± 2.56	98.13 ± 1.11	$\mathbf{99.88 \pm 0.13}$	87.90 ± 1.77	95.30 ± 1.19	99.82 ± 0.27	99.09 ± 0.29
16	27.08 ± 9.00	90.44 ± 1.62	90.87 ± 1.55	67.90 ± 17.85	88.87 ± 3.03	$\mathbf{93.87 \pm 2.27}$	89.67 ± 1.54
17	74.27 ± 10.87	96.87 ± 0.85	94.59 ± 0.67	61.49 ± 16.99	96.09 ± 2.93	98.35 ± 1.17	$\mathbf{98.51 \pm 2.42}$
18	63.04 ± 10.48	$\mathbf{99.40 \pm 0.34}$	99.21 ± 0.15	58.22 ± 12.17	85.21 ± 4.06	95.36 ± 1.25	98.78 ± 0.38
19	57.44 ± 7.63	89.43 ± 1.84	$\mathbf{92.35 \pm 1.15}$	50.08 ± 3.57	86.49 ± 3.93	83.16 ± 1.44	88.57 ± 1.32
20	74.39 ± 1.74	83.14 ± 5.55	91.46 ± 2.18	74.57 ± 4.42	86.21 ± 5.95	86.88 ± 3.90	$\mathbf{96.32 \pm 1.50}$
21	63.57 ± 10.32	91.53 ± 2.88	92.63 ± 5.38	90.67 ± 2.05	88.47 ± 5.21	92.99 ± 2.18	$\mathbf{99.42 \pm 0.42}$
22	86.23 ± 4.55	86.53 ± 5.03	$\mathbf{98.43 \pm 1.67}$	64.00 ± 5.52	95.70 ± 1.43	97.10 ± 2.24	91.33 ± 2.38
OA(%)	76.31 ± 1.34	83.93 ± 0.36	88.80 ± 0.65	73.59 ± 0.44	82.52 ± 2.07	88.84 ± 0.14	$\mathbf{89.57 \pm 0.44}$
AA(%)	65.45 ± 1.75	82.82 ± 0.68	87.46 ± 0.29	62.58 ± 0.55	79.50 ± 1.39	85.39 ± 0.35	$\mathbf{87.95 \pm 0.51}$
$\kappa \times 100$	70.65 ± 1.66	80.10 ± 0.42	86.02 ± 0.77	67.36 ± 0.53	78.37 ± 2.29	85.97 ± 0.19	$\mathbf{86.90 \pm 0.55}$

TABLE VI
CLASSIFICATION RESULTS OF EXPERIMENTS ON THE SALINAS DATASET (TEN LABELS PER CLASS)

Class	HiT	SSFTT	CASST	SC-EADNet	S3Net	C-SS-MTr	Ours
1	74.62 ± 12.86	99.29 ± 1.40	99.94 ± 0.09	$\mathbf{100.00 \pm 0.00}$	99.92 ± 0.11	93.15 ± 2.57	99.78 ± 0.15
2	90.55 ± 11.47	99.98 ± 0.04	99.97 ± 0.06	99.98 ± 0.01	99.28 ± 0.82	94.87 ± 1.34	$\mathbf{100.00 \pm 0.00}$
3	96.01 ± 3.13	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	99.69 ± 0.68	$\mathbf{100.00 \pm 0.00}$	98.30 ± 0.57	$\mathbf{100.00 \pm 0.00}$
4	95.50 ± 1.72	$\mathbf{100.00 \pm 0.00}$	99.83 ± 0.08	$\mathbf{100.00 \pm 0.00}$	97.70 ± 0.82	99.00 ± 0.69	99.01 ± 0.33
5	95.98 ± 4.96	93.28 ± 2.67	90.43 ± 0.44	97.66 ± 1.05	$\mathbf{99.93 \pm 0.08}$	88.62 ± 1.04	98.55 ± 0.25
6	98.60 ± 1.86	99.96 ± 0.08	$\mathbf{100.00 \pm 0.00}$	99.29 ± 1.21	99.97 ± 0.04	94.87 ± 0.98	97.47 ± 0.71
7	92.15 ± 3.32	99.89 ± 0.24	$\mathbf{99.99 \pm 0.01}$	98.92 ± 2.08	99.98 ± 0.04	98.62 ± 0.45	99.99 ± 0.02
8	82.06 ± 8.47	83.57 ± 2.07	83.10 ± 0.83	88.41 ± 4.04	88.37 ± 3.02	76.32 ± 1.91	$\mathbf{92.92 \pm 3.84}$
9	98.35 ± 0.57	99.97 ± 0.05	$\mathbf{100.00 \pm 0.00}$	99.25 ± 0.65	99.84 ± 0.36	99.36 ± 0.29	$\mathbf{100.00 \pm 0.00}$
10	84.59 ± 4.35	97.00 ± 0.77	96.68 ± 0.49	99.44 ± 0.45	95.52 ± 1.15	96.75 ± 1.00	$\mathbf{99.76 \pm 0.11}$
11	82.76 ± 11.17	99.85 ± 0.11	99.72 ± 0.33	99.83 ± 0.19	99.45 ± 0.40	99.22 ± 0.31	$\mathbf{100.00 \pm 0.00}$
12	93.40 ± 4.95	98.07 ± 2.53	97.29 ± 1.99	$\mathbf{100.00 \pm 0.00}$	99.72 ± 0.63	97.87 ± 0.72	99.20 ± 0.81
13	74.97 ± 10.27	96.36 ± 0.82	95.61 ± 1.70	98.87 ± 0.48	$\mathbf{99.98 \pm 0.05}$	97.97 ± 1.30	99.96 ± 0.06
14	88.15 ± 8.03	98.60 ± 0.49	97.66 ± 1.12	$\mathbf{99.96 \pm 0.08}$	99.72 ± 0.28	97.26 ± 0.73	98.27 ± 0.39
15	65.47 ± 6.79	82.78 ± 3.84	81.62 ± 1.87	88.16 ± 5.71	$\mathbf{94.32 \pm 2.68}$	90.16 ± 3.44	91.96 ± 4.99
16	89.35 ± 8.47	94.65 ± 3.12	97.31 ± 0.93	99.83 ± 0.15	97.47 ± 0.22	97.91 ± 0.84	$\mathbf{99.96 \pm 0.03}$
OA(%)	80.82 ± 2.47	93.36 ± 0.51	93.01 ± 0.19	95.59 ± 0.39	96.29 ± 0.32	91.65 ± 0.24	$\mathbf{97.07 \pm 0.09}$
AA(%)	87.66 ± 2.06	96.45 ± 0.62	96.20 ± 0.20	98.08 ± 0.30	98.20 ± 0.10	95.11 ± 0.07	$\mathbf{98.55 \pm 0.13}$
$\kappa \times 100$	78.83 ± 2.72	92.62 ± 0.57	92.22 ± 0.21	95.10 ± 0.43	95.88 ± 0.35	90.74 ± 0.26	$\mathbf{96.74 \pm 0.09}$

12th, 20th, and 21st categories obtained significant results of 4.74%, 7.03%, 4.86%, and 6.43%, respectively. There is a significant improvement, while most of the other poorer performing subclasses have smaller gaps compared with the best performing methods. For the SA dataset, as shown in Table VI, the classification accuracy of most subclasses under these methods is very high, leading to limited room for improvements. However, our method still performs best with the classification accuracy for most subclasses close to 100%, especially for the eighth subclass, which has a significant improvement of 4.51%, verifying the effectiveness of our proposed method again.

We also validated the performance of each method with different training sample sizes. As shown in Fig. 9, the experiments indicate that as the sample size increases, the classification accuracy of each model improves to varying

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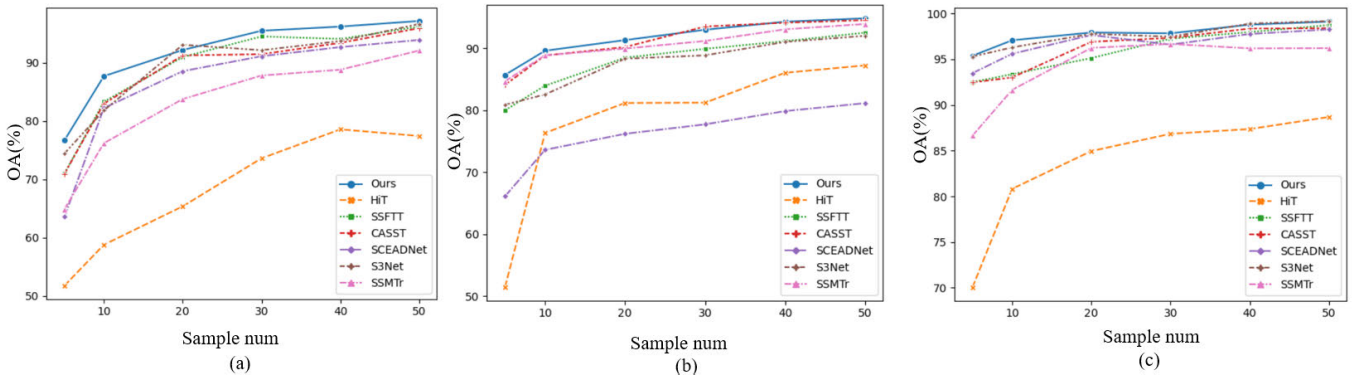


Fig. 9. Overall classification accuracy of various algorithms under different training sample sizes. (a) IP. (b) WHHH. (c) SA.

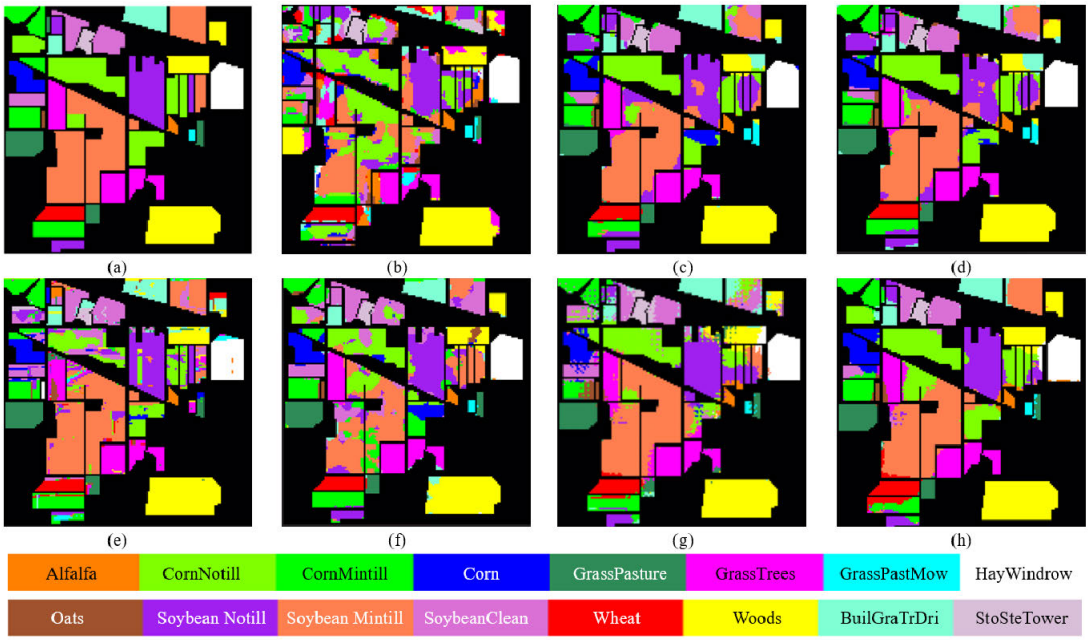


Fig. 10. Visual classification results of Indian Pines from compared methods. (a) Ground truth. (b) HiT. (c) SSFTT. (d) CASST. (e) SC-EADNet. (f) S3Net. (g) C-SS-MTr. (h) Ours.

degrees. Notably, the proposed model demonstrates competitive performance across different training sample sizes on the IP, WHHH, and SA datasets. This experiment further confirms the effectiveness of the proposed method.

2) Visual Evaluation: To visualize the performance of different algorithms on the three datasets and to compare the classification results of our method, we present the final classification map for the IP, WHHH, and SA datasets in Figs. 10–12, respectively. Compared with the ground truth, our proposed model exhibits the fewest noise points and the best classification results in terms of visual quality.

Specifically, the HiT model demonstrates lower performance compared with other algorithms across all three datasets, particularly showing more noise points in subclasses with smaller area in the IP and WHHH datasets, and in the Bricks subclass of the SA dataset. As detailed results indicated in Table IV, the OA of the HiT model is 28.88%, 13.26%, and 16.25% lower than our method for the IP, WHHH, and SA datasets, respectively. Furthermore, Fig. 11 reveals that the S3Net method has misclassification noise points in the Cotton category of the WHHH dataset. Correspondingly, Table V shows that this method has the lowest accuracy in this category, with a difference of 7.33% compared with our method. In addition, the SC-EADNet method displays misclassifications within similar land cover clusters across all the subclasses in the WHHH dataset. Table V confirms that SC-EADNet has the worst performance, with an OA 15.98% lower than our method, and other models also exhibit varying degrees of misclassification noise points. Compared with these methods, our method demonstrates superior classification accuracy across the IP, WHHH, and SA datasets, particularly in regions with high land cover density within a unit area (e.g., the upper left part of IP, the top and bottom right parts of WHHH, and the right and bottom parts of SA). These results indicate that our model has strong classification capabilities in scenarios with limited training samples.

C. Ablation Study

To further validate the effectiveness of the proposed model framework, the FNSM, and the HENSM strategies, we designed and conducted ablation studies to explore

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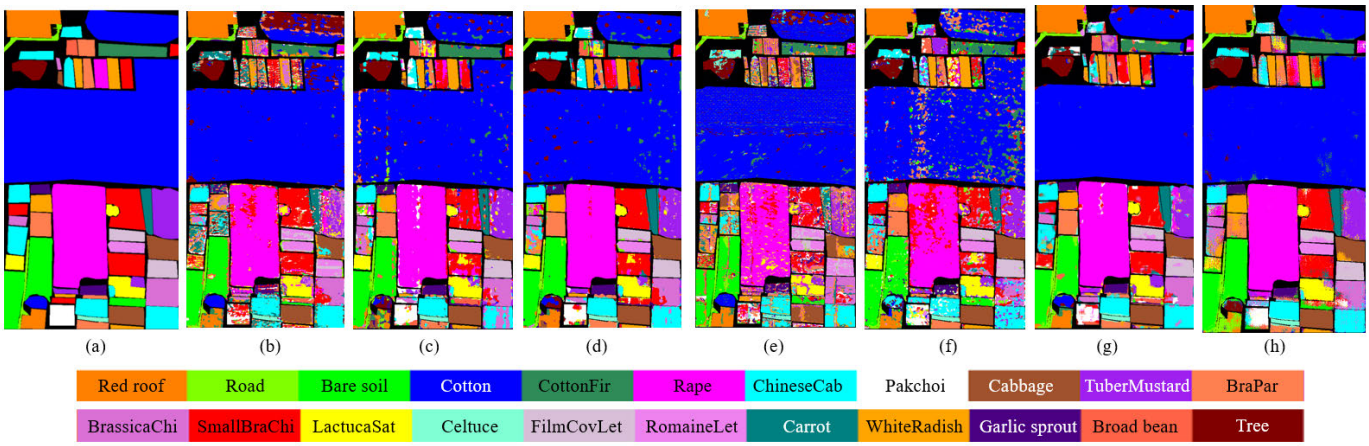


Fig. 11. Visual classification results of WHU-Hi-HongHu from compared methods. (a) Ground truth. (b) HiT. (c) SSFTT. (d) CASST. (e) SC-EADNet. (f) S3Net. (g) C-SS-MTr. (h) Ours.

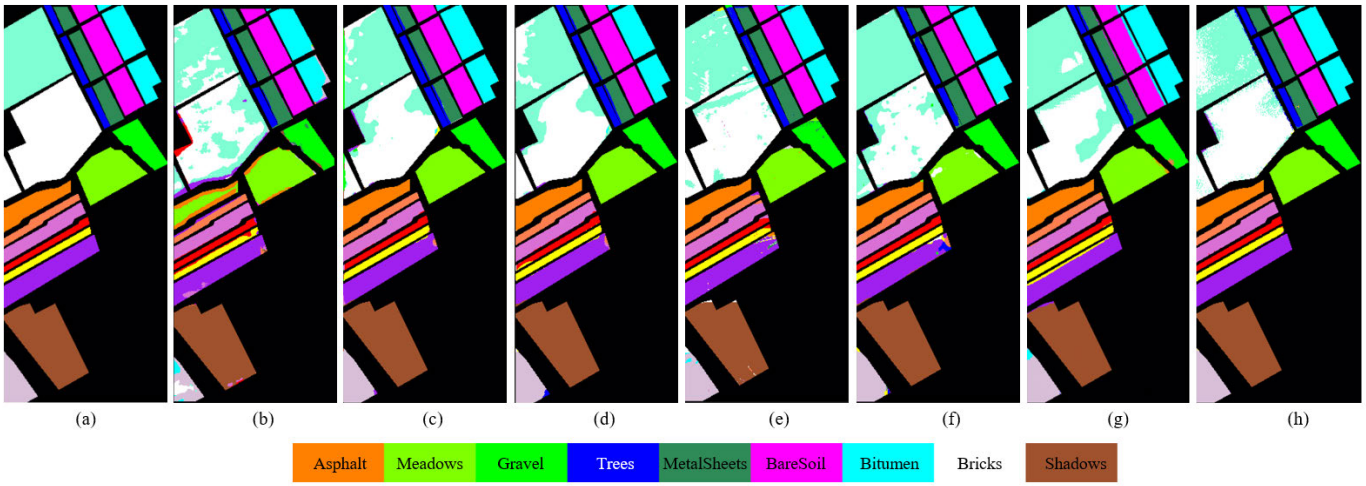


Fig. 12. Visual classification results of Salinas from compared methods. (a) Ground truth. (b) HiT. (c) SSFTT. (d) CASST. (e) SC-EADNet. (f) S3Net. (g) C-SS-MTr. (h) Ours.

TABLE VII

CLASSIFICATION PERFORMANCE OF THE PROPOSED CASST METHOD WITH DIFFERENT ABLATION STRATEGIES ON THE THREE HSI DATASETS

Index	RNS	FNSM	HENSM	Indian Pines			WHU-HI-HONGHU			Salinas		
				OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
1				85.13 ± 0.51	92.27 ± 0.48	83.21 ± 0.59	88.83 ± 0.75	86.94 ± 0.67	85.98 ± 0.91	95.98 ± 0.87	98.08 ± 0.39	95.52 ± 0.97
2	✓			85.35 ± 0.59	92.28 ± 0.34	83.44 ± 0.65	89.47 ± 1.21	87.31 ± 1.11	86.78 ± 1.49	96.06 ± 0.58	98.48 ± 0.22	96.61 ± 0.31
3		✓		86.42 ± 0.40	93.12 ± 0.34	84.67 ± 0.45	89.13 ± 0.42	87.01 ± 0.74	86.36 ± 0.53	96.83 ± 0.66	98.49 ± 0.41	96.47 ± 0.74
4		✓	✓	87.67 ± 0.95	92.71 ± 0.71	86.06 ± 1.05	89.57 ± 0.44	87.95 ± 0.51	86.90 ± 0.55	97.07 ± 0.09	98.55 ± 0.13	96.74 ± 0.09

· RNS: Random-Negative Sampling. FNSM: False-Negative Samples Mitigation. HENSM : Hard-Easy Negative Samples Mixing Mining.

their performance and innovation. The traditional contrastive-learning-based HIS classification task typically uses RNS to construct a batch of data. In contrast, we propose FNSM and HENSM to eliminate optimization inconsistencies caused by such random sampling. To rigorously verify the effectiveness of our proposed strategies, we designed four ablation experimental scenarios, maintaining the same network framework, hyperparameters, and dataset partitioning while modifying the respective modules. Specifically, the scenarios include the below.

1) *Baseline Model:* Only conduct the classification task, excluding the contrastive learning task.
2) *RNS:* Incorporates contrastive learning to assist the classification task without additional negative sample mining strategy.
3) *False-Negative Samples' Mitigation:* Builds on the contrastive-learning-assisted classification task by introducing the FNSM strategy guided by pseudolabels during negative sample mining.
4) *Combined Strategy (FNSM + HENSM):* Integrates both the false-negative mitigation and hard–easy negative sample mining strategies in the contrastive-learning-assisted classification task.

Table VII presents the classification performance of the proposed method on three HSI datasets (IP, WHHH, and SA) under different ablation strategies. The baseline model achieved moderate performance across all three datasets, serving as a benchmark for further comparison. It is observed that the introduction of the contrastive-learning-assisted classification task, without any negative sample mining strategy

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(i.e., the RNS-based method), slightly improves the baseline model. The OA on the IP, WHHH, and SA datasets increased marginally to 85.35%, 89.47%, and 96.06%, respectively, representing improvements of 0.22%, 0.64%, and 0.08%. Similarly, AA and κ metrics showed slight enhancements, indicating limited benefits from RNS.

Considering negative sample mining, the implementation of the pseudolabel guided FNSM strategy yields more significant improvements. The OA values for the IP, WHHH, and SA datasets increased to 86.42%, 89.13%, and 96.83%, respectively, reflecting further increases of 1.29%, 0.3%, and 0.85% compared with the baseline model with the RNS strategy. The AA and κ metrics also demonstrated corresponding improvements, suggesting that addressing the errors introduced by RNS and reducing false-negatives can effectively enhance model performance, leading to more consistent and reliable classification.

Finally, the model combining both the FNSM and HENSM strategies achieved the highest performance across all the datasets. The OA improved by 2.54%, 0.74%, and 1.09% for the IP, WHHH, and SA datasets, respectively, compared with the baseline model. The AA and κ values also reached their highest levels. These results demonstrate the complementary advantages of the two strategies, with ENSM further refining the model's ability to handle negative samples by focusing on the hard–easy sample mix ratio, thereby providing more robust and accurate classification.

IV. Conclusion

This study proposed a novel hyperspectral contrastive learning classification framework that can flexibly support informative negative sample mining, rather than the RNS methods used in conventional contrastive learning algorithms. Based on this framework, we further presented two negative sample mining strategies guided by pseudolabels generated by the classifier: the FNSM and the HENSM strategies.

To evaluate the accuracy of the proposed contrastive-learning-based HSI classification framework, as well as the effectiveness of the FNSM and HENSM strategies, we comprehensively conducted a series of comparison experiments and the ablation study. The results based on comparison experiments demonstrated that our proposed method further improves the HIS classification performance, with OA, AA, and κ all achieving the best results compared with the second-best method, with OA improvements of 4.37%, 0.73%, and 0.78% on the IP, PU, and WHHH datasets, respectively. In addition, the ablation study reveals that our classification framework using both the FNSM and HENSM strategies achieved the highest performance across all the datasets compared with the baseline model, with OA improving by 2.54%, 0.74%, and 1.09% for the three datasets, respectively. This validates the effectiveness of the proposed FNSM and HENSM strategies in addressing the limitations of traditional RNS methods, leading to superior classification performance across multiple datasets. This should be attributed to the FNSM strategy is proposed on the basis of enhancing the consistency between the optimization direction of contrastive learning and the classification task, while the HENSM strategy is designed to improve the classification performance by controlling the difficulty of negative samples, thus reducing the learning complexity of contrastive learning. These findings underscore the importance of targeted negative sample management in improving the overall accuracy of HSI classification models, suggesting that negative sample mining really matters, with the effective negative sample mining strategies significantly enhancing the learning outcomes of HSI contrastive learning tasks, and result in improved final classification accuracy. Furthermore, more experiments in our future research will investigate the potential of negative samples mining in HSI classification tasks based on contrastive learning. We will delve into exploring optimal strategies for negative samples mining and pairing positive and negative samples, and further connect it with the effective models to ultimately serve for HSI classification tasks in various application fields.

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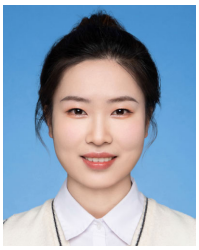
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